

```

        newsArticles = ....
    }

    fun getNewsArticles(): LiveData<List<NewsArticles>> {
        return newsArticles
    }
}

```

Now, let's look at our UI controller (Activity and Fragment):

```

class NewsActivity: AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_news)
        // Get ViewModel
        val newsViewModel = ViewModelProviders.of(this).
            get(NewsViewModel::class.java)

        // Observing for data change
        newsViewModel.getNewsArticles().observe(this,
            Observer<List<NewsArticles>> {
                // Update the UI
            })
    }
}

```

In the Activity or Fragment we get an instance of the ViewModel using *ViewModelProviders* and then get the data using the 'getters' that we defined. As we are

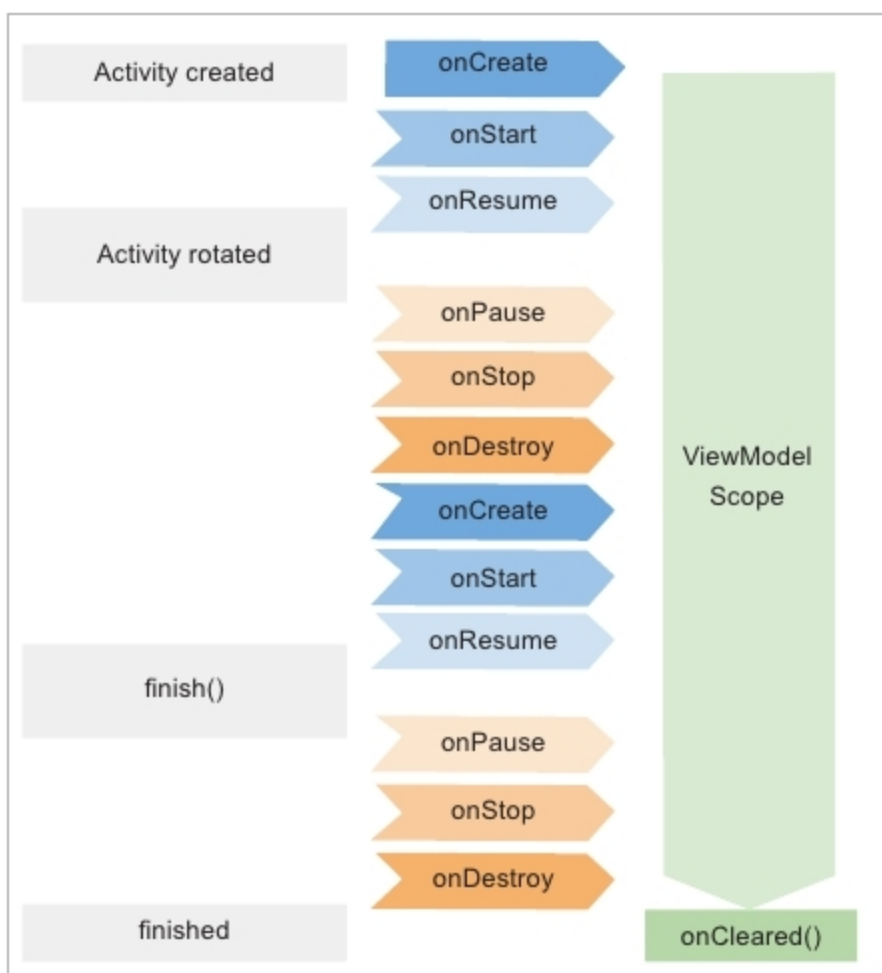


Figure 4: ViewModel

returning data wrapped in LiveData, we will be able to observe data changes.

Behind the scenes: ViewModel is created when the Activity is created and is destroyed only when the Activity is concluded. Finally, when the owner Activity is finished, the framework automatically calls ViewModel's *onCleared()* method so that it can clean up resources.

Listed below are some of the advantages of ViewModel:

- It is designed to hold and manage UI related data
- Survives configuration changes
- It is life cycle-aware
- It will often store LiveData objects
- Helps to communicate between Activity and Fragment
- Eliminates 'God Activities'

Room

Room is a persistence library designed to help you save your app data in the SQLite database that ships with every Android version.

Its features include the following:

- It provides local data persistence
- An abstraction layer over the existing SQLite database
- Checks SQL queries at compile time
- Has the ability to observe for changes in the database using LiveData
- Eliminates the boilerplate code
- Plays well with LiveData and RxJava

Anatomy of Room: There are various components of the Room persistence library and they include the following.

Entity

- For each entity, a database table is created.
- It represents a class, which holds a database row.

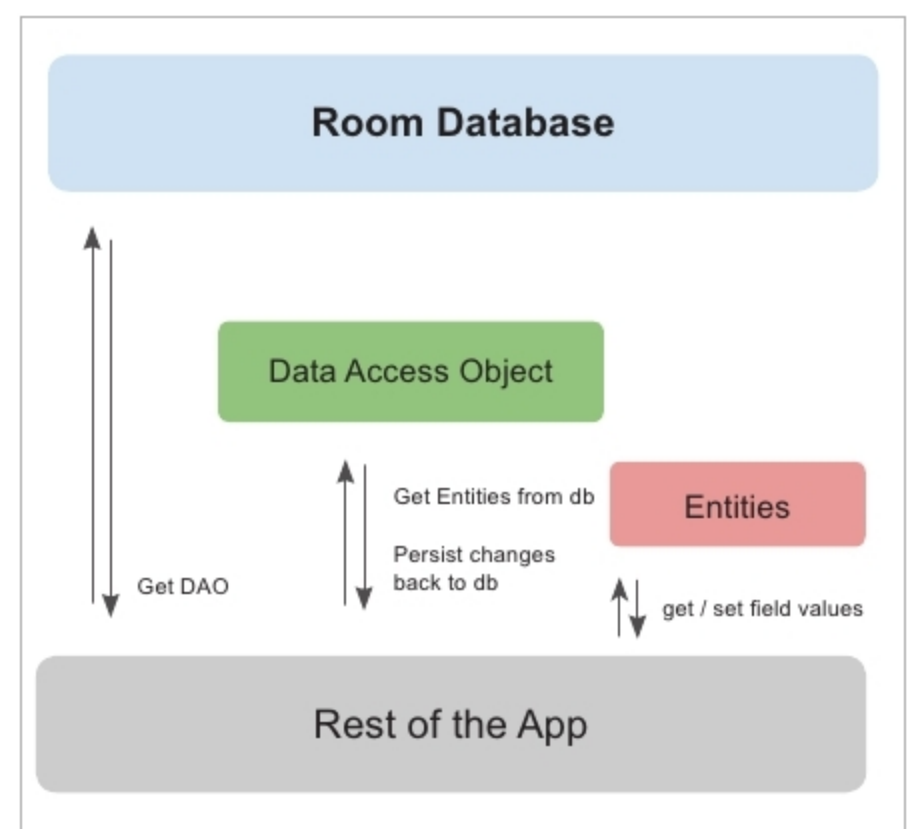


Figure 5: Room architecture